Analytical Chemistry Cumulative Exam
October 18, 2024
Topic: Statistics and Error Propagation

You must show your working; no credit will be given for answers without working. Partial credit will be given for each step of the solution – A correctly set up problem calculator errors/miscopying numbers can still pass this cume!

Statistical tables and formulae are available at the end of the exam.

You should use the appropriate number of significant figures in all your answers.

Question 1 – This question assesses your ability to choose the appropriate statistical test for a given scenario – *It is not testing your ability to apply the test*. For each of the scenarios listed below identify the appropriate statistical test by providing the equation for the test. While example data are given, for this question you do not need to perform any calculations or apply the statistical test.

a) Two analytical methods are developed to measure the amount of arsenic in drinking water. Samples of water from different wells are analyzed using two analytical methods. The results of the two methods are below. What is an appropriate statistical test to determine if the methods differ significantly in the concentration of arsenic determined?

	Method 1 (As / ppb)	Method 2 (As / ppb)
Well 1	5.21	5.93
Well 2	14.2	12.7
Well 3	…	…
…		
Well N	…	…

b) The protein content from two different batches of wheat grown in different soils was measured spectroscopically. What is an appropriate statistical test to determine if the protein content of the two samples is different?

Batches 1 (protein %)	Sample 2 (protein %)
13.1	10.4
14.2	15.7
11.2	18.2
…	…
15.1	13.2

c) A student in the lab claims that a spectrometer is the source of high variability in their measurements. You test the spectrometer with a standard that you prepared today and compare it to measurements of a standard you took last week (the concentrations of the standards are different). What statistical test should you perform on your data sets to determine if the variability of the instrument has changed?

This week (mM analyte)	Last week (mM analyte)
22.3	10.4
22.8	10.8
21.9	11.1
…	…
22.0	10.7

Question 2 The concentration of F^- in solution (in mg/L) is related to the potential (*E*) of an ion-selective electrode by Conc. F^- (mg/L) = $10^{(K-E)/59\ mV}$ where *K* is a constant that depends on the electrode properties.

Your electrode has a calibration coefficient of *K* = 109±2 mV. You use it to measure a tap-water solution and get a measurement of *E* = 99±3 mV. What is the concentration of fluoride in the solution and the absolute error in this value? (use error propagation where appropriate).

Answer format: Your answer should be in the form 12.345±0.6789 or 1.23×10^4±5.67 (where you should choose appropriate significant figures for both the answer and the error and you should choose appropriate units). *Please work the question in a step-by-step manner and write down all the steps of your calculation so that you can get partial credit.*

Question 3

You are comparing the precision of two different assays for Mg^{2+} in seawater. The values of the magnesium as determined by the two methods are listed in the tables below. At the 95% confidence level, is there a significant difference between <u>the precision</u> of the two methods? (show your working)

Your answer should contain 3 parts: 1) your calculated value of the test statistic 2) the critical value that you are comparing to. 3) "yes the precisions are statistically different" or "no the precisions are not statistically different"

(question continues on following page)

For your convenience, the means and standard deviation of the two methods are provided in the table.

	Method 1	**Method 2**
	(g/L)	(g/L)
	11.8	16.4
	11.8	14.7
	12.0	16.1
	12.2	18.7
	13.1	16.3
	11.8	
	11.5	
	11.8	
Mean	**12.0**	**16.4**
Std. dev.	**0.4_9**	**1.4_4**

Constants, Equations, and Tables

$$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$E = x_i - x_t$$

$$E_r = \frac{x_i - x_t}{x_t} \times 100\%$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

$$s = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N-1}} = \sqrt{\frac{\sum_{i=1}^{N} x_i^2 - \frac{\left(\sum_{i=1}^{N} x_i\right)^2}{N}}{N-1}}$$

$$s_m = \frac{s}{\sqrt{N}}$$

$$s_{pooled} = \sqrt{\frac{\sum_{i=1}^{N_1} (x_i - \overline{x_1})^2 + \sum_{j=1}^{N_2} (x_j - \overline{x_2})^2 + \sum_{k=1}^{N_3} (x_k - \overline{x_3})^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$s_{pooled} = \sqrt{\frac{(N_1 - 1)\, s_1^2 + (N_2 - 1)\, s_2^2 + (N_3 - 1)\, s_3^2 + \ldots}{N_1 + N_2 + N_3 + \ldots - N_t}}$$

$$y = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}} = \frac{e^{-z^2/2}}{\sigma\sqrt{2\pi}}$$

$$z = \frac{(x - \mu)}{\sigma}$$

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$$\mu = \bar{x} \pm \frac{z\,\sigma}{\sqrt{N}}$$

$$\mu = \bar{x} \pm \frac{t\,s}{\sqrt{N}}$$

$$z = \left| \frac{\bar{x} - \mu_o}{\sigma / \sqrt{N}} \right|$$

$$t = \left| \frac{\bar{x} - \mu_o}{s / \sqrt{N}} \right|$$

$$t = \left| \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled} \sqrt{\frac{N_1 + N_2}{N_1 N_2}}} \right|$$

$$z = \left| \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{N_1 + N_2}{N_1 N_2}}} \right|$$

$$s_d = \sqrt{\frac{\sum \left(d_i - \bar{d}\right)^2}{N - 1}}$$

$$t = \frac{|\bar{d}|}{s_d / \sqrt{N}}$$

$$F = \frac{s_1^2}{s_2^2}$$

$$G_{calc} = \frac{|\text{questionable value} - \bar{x}|}{s}$$

$$RSD = \frac{s}{\bar{x}}$$

$$CV = \frac{s}{\bar{x}} \times 100\%$$

$$w = \left(x_{largest} - x_{smallest}\right)$$

	Function	Uncertainty
+/-	$y = x_1(\pm e_{x_1}) + x_2(\pm e_{x_2}) - x_3(\pm e_{x_3})$	$e_y = \sqrt{e_{x_1}^2 + e_{x_2}^2 + e_{x_3}^2}$
×/÷	$y = \dfrac{x_1(\pm e_{x_1}) \times x_2(\pm e_{x_2})}{x_3(\pm e_{x_3})}$	$\dfrac{e_y}{y} = \sqrt{\left(\dfrac{e_{x_1}}{x_1}\right)^2 + \left(\dfrac{e_{x_2}}{x_2}\right)^2 + \left(\dfrac{e_{x_3}}{x_3}\right)^2}$
	where $\%e_{x_1} = 100\% \dfrac{e_{x_1}}{x_1}, \%e_{x_2} = 100\% \dfrac{e_{x_2}}{x_2}$, $\%e_{x_3} = 100\% \dfrac{e_{x_3}}{x_3}, \%e_y = 100\% \dfrac{e_y}{y}$	or $\%e_y = \sqrt{(\%e_{x_1})^2 + (\%e_{x_2})^2 + (\%e_{x_3})^2}$
exponentiation	$y = x(\pm e_x)^a$	$\dfrac{e_y}{y} = a\,\dfrac{e_x}{x}$
		or $\%e_y = a\ \%e_x$ where $\%e_y = 100\% \dfrac{e_y}{y}, \%e_x = 100\% \dfrac{e_x}{x}$
logarithm	$y = \log_{10}(x(\pm e_x))$	$e_y = \dfrac{1}{\ln 10}\dfrac{e_x}{x} = 0.43429\ \dfrac{e_x}{x}$
	$y = \ln\left(x(\pm e_x)\right)$	$e_y = \dfrac{e_x}{x}$
antilogarithm	$y = \text{antilog}_{10}(x(\pm e_x)) = 10^{x(\pm e_x)}$	$\dfrac{e_y}{y} = (\ln 10)\, e_x \approx 2.3026\, e_x$

TABLE 4-5 Critical values of *G* for rejection of outlier

Number of observations	*G* (95% confidence)
4	1.463
5	1.672
6	1.822
7	1.938
8	2.032
9	2.110
10	2.176
11	2.234
12	2.285
15	2.409
20	2.557

$G_{calculated}$ = |*questionable value − mean*|/*s*. *If* $G_{calculated} > G_{table}$, *the value in question can be rejected with 95% confidence. Values in this table are for a one-tailed test, as recommended by ASTM.*

TABLE 4-4 Values of Student's *t*

Degrees of freedom	Confidence level (%) 50	90	95	98	99	99.5	99.9
1	1.000	6.314	12.706	31.821	63.656	127.321	636.578
2	0.816	2.920	4.303	6.965	9.925	14.089	31.598
3	0.765	2.353	3.182	4.541	5.841	7.453	12.924
4	0.741	2.132	2.776	3.747	4.604	5.598	8.610
5	0.727	2.015	2.571	3.365	4.032	4.773	6.869
6	0.718	1.943	2.447	3.143	3.707	4.317	5.959
7	0.711	1.895	2.365	2.998	3.500	4.029	5.408
8	0.706	1.860	2.306	2.896	3.355	3.832	5.041
9	0.703	1.833	2.262	2.821	3.250	3.690	4.781
10	0.700	1.812	2.228	2.764	3.169	3.581	4.587
15	0.691	1.753	2.131	2.602	2.947	3.252	4.073
20	0.687	1.725	2.086	2.528	2.845	3.153	3.850
25	0.684	1.708	2.060	2.485	2.787	3.078	3.725
30	0.683	1.697	2.042	2.457	2.750	3.030	3.646
40	0.681	1.684	2.021	2.423	2.704	2.971	3.551
60	0.679	1.671	2.000	2.390	2.660	2.915	3.460
120	0.677	1.658	1.980	2.358	2.617	2.860	3.373
∞	0.674	1.645	1.960	2.326	2.576	2.807	3.291

In calculating confidence intervals, σ may be substituted for s in Equation 4-7 if you have a great deal of experience with a particular method and have therefore determined its "true" population standard deviation. If σ is used instead of s, the value of t to use in Equation 4-7 comes from the bottom row of this table.

Values of t in this table apply to two-tailed tests illustrated in Figure 4-9a. The 95% confidence level specifies the regions containing 2.5% of the area in each wing of the curve. For a one-tailed test, we use values of t listed for 90% confidence. Each wing outside of t for 90% confidence contains 5% of the area of the curve.

TABLE 4-4 Critical values of $F = s_1^2/s_2^2$ at 95% confidence level

Degrees of freedom for s_2	Degrees of freedom for s_1: 2	3	4	5	6	7	8	9	10	12	15	20	30	∞
2	19.0	19.2	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5
3	9.55	9.28	9.12	9.01	8.94	8.89	8.84	8.81	8.79	8.74	8.70	8.66	8.62	8.53
4	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.75	5.63
5	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.50	4.36
6	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.81	3.67
7	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.58	3.51	3.44	3.38	3.23
8	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.08	2.93
9	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.86	2.71
10	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.84	2.77	2.70	2.54
11	3.98	3.59	3.36	3.20	3.10	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.57	2.40
12	3.88	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.47	2.30
13	3.81	3.41	3.18	3.02	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.38	2.21
14	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.31	2.13
15	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.25	2.07
16	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.19	2.01
17	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.15	1.96
18	3.56	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.11	1.92
19	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.07	1.88
20	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.04	1.84
30	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.84	1.62
∞	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.46	1.00

Critical values of F for a one-tailed test of the hypothesis that $s_1 > s_2$. There is a 5% probability of observing F above the tabulated value.

You can compute F for a chosen level of confidence with the Excel function FINV(probability,deg_freedom1,deg_freedom2). The statement "=FINV(0.05,7,6)" reproduces the value F = 4.21 in this table. The statement "=FINV(0.1,7,6)" gives F = 3.01 for 90% confidence.

Harris, *Quantitative Chemical Analysis*, 8e
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